PROJECT GUIDE

HERE & THERE
how the app works

An illustrated tour for anyone new to the project: what the app does, how a request travels through it, and the reasoning behind each rule.

HERE

What can I do in the next few hours?

THERE

Who can I travel with?

Rule values (weights, limits, days) are quoted from the code as of 2026-10-26. Each section names the file that owns the rule, so the current value can always be checked.

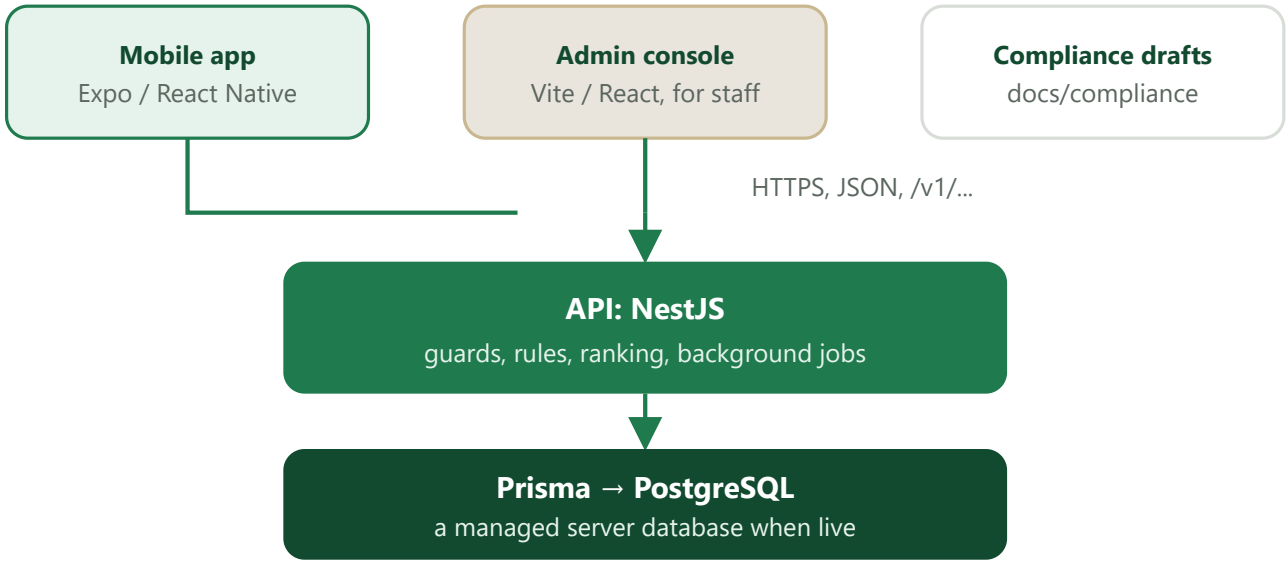
1 The product in one minute

One account, two modes. People can follow each other; people who follow each other are **buddies**. Both modes show you what your circle is doing before they show you strangers.

HERE
Short-notice plans near you.
Coffee in 90 minutes, a pickup game tonight.

THERE
Trips and travel companions.
A 5-day trip to Lisbon next month.

2 The pieces and how they connect



Part	Folder	Notes
API	`services/api`	All rules live here; the apps stay thin
Mobile app	`apps/mobile`	Signed-out users only see the sign-in screens
Admin console	`apps/admin`	Reports, verifications, no-shows, security
Compliance drafts	`docs/compliance`	Privacy notice, DPIA, retention. Drafts, need lawyer review

3 Life of a request

Example: you open Home.

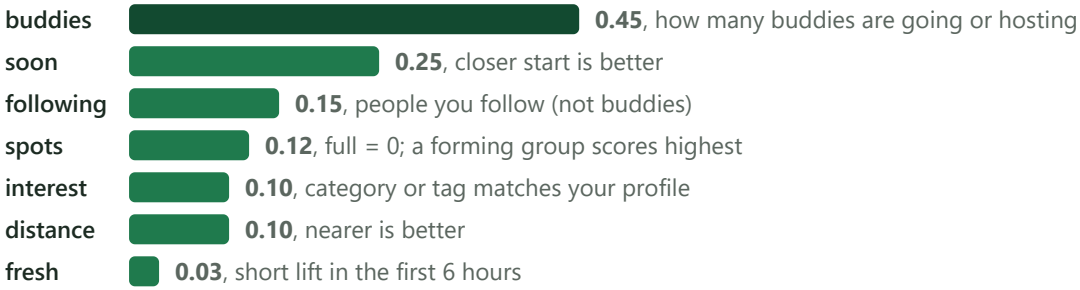
1. **App** calls `GET /v1/activities?lat=..&lng=..` with your session token and a device id.
2. **Guard** checks the token: valid, not expired, not an unfinished two-step pass, and the password version still matches. Changed your password elsewhere? This token is refused.
3. **Service** builds the candidate list, scores it and picks the best page (section 4).
4. **Response** carries the items, a `nextCursor` and an `ETag`.

5. **App** shows placeholders while loading, then the cards. Reaching the bottom sends the cursor to fetch the next page.
6. Later refreshes send `If-None-Match`. If nothing changed the API answers **304** with no body (section 7).

4 The Home feed: scoring and Top-K

1 Candidates up to 500 upcoming, visible to you › **2 Filter** radius, "has spots" › **3 Score** 0 to 1 each › **4 Top-K** best limit+1 via heap › **5 Page** send limit + cursor

The extra (limit + 1th) item only tells us there is a next page. Each activity's final score is a **weighted average** of seven signals, each between 0 and 1:



FAIRNESS RULE

Missing signals are **left out, not counted as zero**. No interests in your profile, or no shared location? Those weights are removed from both the sum and the divisor, so people who share less are not penalised.

`score = Σ(weight × signal) ÷ Σ(weights that apply)`

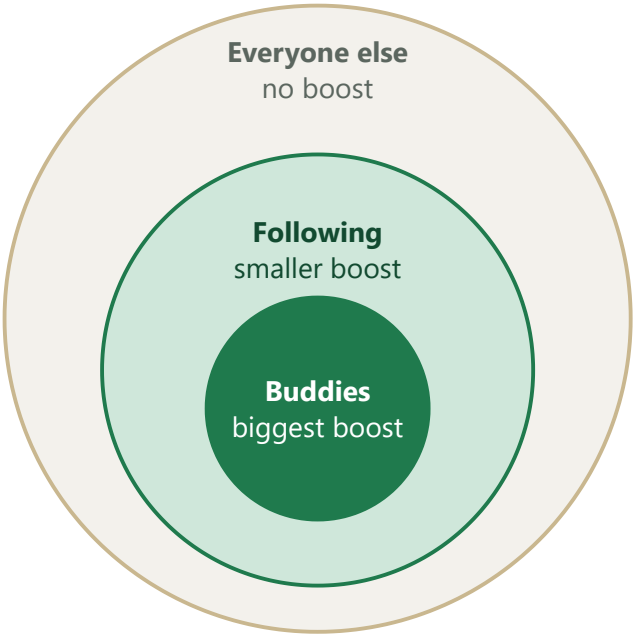
Top-K with a heap

Sorting 500 items to show 20 is wasteful. We keep a **min-heap** of the best K items seen so far. Its root is the weakest survivor, so a new item only has to beat the root to get in: cost O(n log K) instead of O(n log n).

Heap not full? add the item | **Better than the root?** replace root, re-balance | **Otherwise** ignore it

Ties are broken by activity id, so the order never wobbles between requests. The **cursor** stores the time of the first page, the last score and the last id; page 2 is scored on the *same clock* as page 1, so scores do not drift.

5 Circle tiers: how a buddy adds weight



Tier	Who
Buddies	you follow them **and** they follow you back
Following	you follow them, no follow back
Everyone else	including people who only follow you

Only the host and **approved** members count, and each person once.

A count becomes a signal with diminishing returns, `crowd(n) = 1 - 0.4ⁿ`:

Going	0	1	2	3
crowd	0	0.60	0.84	0.94

The first friend matters most; it never passes 1.

Worked example

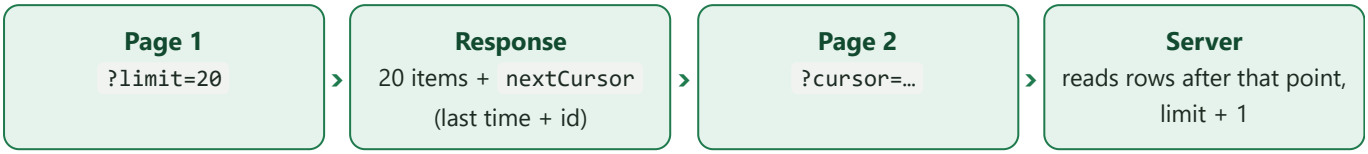
Four identical coffee activities starting in 90 minutes. Only the host differs.

Host	Relationship	Contribution to score sum
Nina	buddy	0.45 × 0.60 = **0.270**
Omar	you follow him	0.15 × 0.60 = **0.090**
Paolo	only follows you	0
Quinn	stranger	0

Order: **Nina, Omar, then Paolo and Quinn** (tie broken by id). The whole "starts very soon" signal adds at most 0.25, so **one buddy going outranks even an activity that starts right now**. That is the intent. Cards explain the order with a small green line: "1 buddy going".

6 Paging: how long lists load

Why not "page 2, page 3"? Skipping rows gets slower the deeper you go and shows duplicates or gaps when rows are added in between. We use **keyset (cursor) paging**.

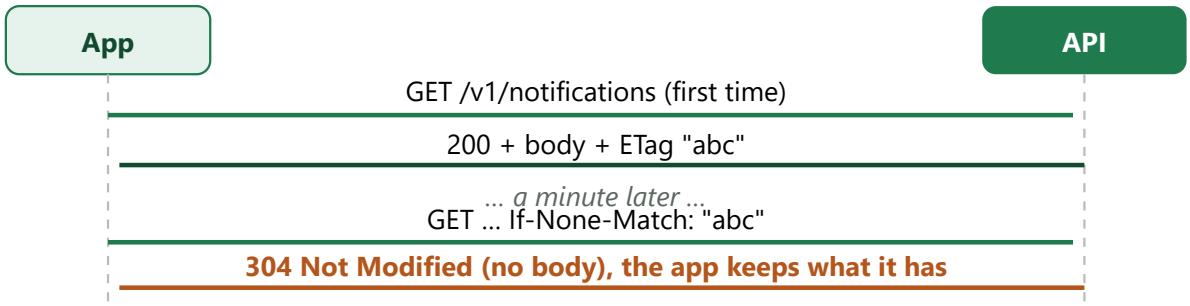


- Cost on page 500 equals cost on page 1.
- `limit` is clamped between 1 and 100. A bad cursor returns `INVALID_CURSOR`.
- On the phone, a refresh re-fetches as many rows as are already loaded, so you are not thrown back to the top.
- Paged lists: favorites, my activities, my trips, alerts, connections, conversations, trip search.

Code: `services/api/src/common/paging.ts` and `apps/mobile/src/lib/paged.tsx`

7 Saving data: ETag and the 304 reply

Polling ("anything new?" on a timer) costs data and battery. Badges, feeds and inbox now check about once a minute, and unchanged answers are skipped:



> **DEBUGGING GOTCHA**
> Browsers add `Cache-Control: no-cache` to such requests and a framework default can turn a match into a 200. A small middleware in `main.ts` strips those headers on GET requests that carry `If-None-Match`.

8 Joining an activity and the waiting list



Greedy, first in line

- Walk the line from the front (oldest first, id as tie-break) and take the first person who can still be let in.
- Skipped but **keeping their place**: inactive accounts and people the host has blocked.

- Read in batches of 25, so a long run of skipped people cannot block those behind them.
- Placement happens in **one database transaction**: capacity re-checked, request still waiting, membership created. The last place taken makes the activity `full`.
- The person is told "A place opened up. You're in!" and an audit row is written. The loop repeats (max 20 times) if several places opened.

9 THERE: how trips are ranked

Trips use a **points score** because a search has many optional preferences (destination, dates, budget, tags, origin city). Everything is a **preference, never an exclusion**: a trip from another origin city only loses a few points.

Buddies going	People you follow going	Everyone else
+30 points max	**+10** points max	**0**

`score += 30 × crowd(buddies) + 10 × crowd(follows)`, using the same `circleOf`, `tierCounts` and `crowd` as HERE. Paging also matches the feed: the same Top-K heap and a last-score-and-id cursor, so nothing is duplicated or skipped between pages. Up to 500 candidate trips are scored per request.

10 Reliability levels (no-shows)

Goal: honest information about who turns up, **without public shaming**. The app shows only one level and one plain line; detail lives in the person's own data download and the admin console.

New not enough history › **Reliable** 3+ kept plans › **Very reliable** 10+ kept plans › **Needs care** worrying recent record; shown to verified people

How the level is computed

1. **Facts** are recorded: kept, no-show, late cancel, host late cancel, upheld report.
2. Older facts count less: **half-life 180 days**, nothing older than 730 days counts.
3. At most **2 kept plans per day** count, so a level cannot be farmed in an afternoon.
4. The rules in `reliability/reliability.ts` turn the weighted facts into a level, re-checked only when something is new.

How a no-show is recorded

Activity ends host marks who did not come (until end + 48 h) › **Member told** can dispute within 7 days › **Moderator decides** admin "No-shows" › **Upheld** counts / **Rejected** removed

Plan status, reminders, drafts and appeals

Status	When
published	open, fewer than 2 people going
confirmed	2+ people going (host counts); everyone going is told once
full	no place left (beats confirmed)

Status moves back automatically when people leave. A **trip** is confirmed when the group reaches its minimum **and** every member confirmed the final dates.

- **"Starting soon" reminder:** a timer finds activities starting within 60 minutes, claims each one so it can never go out twice, and notifies everyone going (not people who joined in the last 15 minutes).
- **Content checks:** swear words in English and Italian (even sh1t, accents, stretched letters), shouting, spam, too many links, scam wording, requests for money. The person sees a short reason and edits the text.
- **Drafts:** half-finished forms saved privately to the account (max 10 per kind, deleted after 90 days); nothing is checked until the draft is published.

Appeal about	Who asks	Who decides
Rejected identity check	the person	trust team
Suspension	the person (email + password)	moderator
Listing removed by staff	its host	moderator
Declined join request / trip application	the person	host or organizer

Each decision can be appealed **once**, and a note is always written back to the person.

11 THERE extras: plan, travel buddies, link safety

Shared plan. Trip members (only) share a day-by-day itinerary and budget notes. Notes are per-person estimates, totalled and compared in plain words with the announced budget ("below", "within", "above"). The app never holds money, and says so. Finished or cancelled trips are read-only.

Trip status: `published` → `full` → `planning` (minimum reached) → `confirmed` (dates agreed) → `completed` (day after last day) or `cancelled`.

Travel buddies: opt-in matching

1	2	3	4
Verified adult says where and when	Sees up to 20 opted-in travelers with overlapping dates	Answers "I would like to chat" or "Not now"	Both said yes: both notified, private 2-person chat opens

> **PRIVACY BY DESIGN**
> A "yes" tells the other person **nothing**: the reply is identical whether or not they had said yes. Contact details are never shown. Either person can end the chat; a block, suspension or account deletion closes it too.

Link and money safety

Allowed
One ordinary web address per message.

Refused
Shortened links, bare IP addresses, look-alike names, `name@host` addresses, risky endings, bank/payment names on the wrong site; requests for money, IBANs, card numbers, wallet addresses.

In trip chat, any message with a link shows a caution, and tapping a link shows a notice first.

Date flexibility: exact, ±1 day, 3 days, 7 days, or anytime that month. The owner chose on 2026-10-26 to keep ±1 day as a deliberate extension.

12 Safety, reports and moderation

Blocking is two-way and works inside group chats: blocked people do not see each other's messages, member lists or alerts, and cannot follow or match.

Report	Urgent first	Queue	Action
user, activity, trip or message; sheet shows "In danger? Call 112"	immediate danger alerts every moderator at once	urgent, overdue, under-18, oldest	hide, warn, suspend, remove, resolve

- **Answer times:** urgent 1 hour, others 24 hours. A late report is **escalated once** to trust leads and administrators. Coverage: every day 08:00 to 22:00 Milan time.
- **Moderators see:** who the report is about, the message with surrounding context, a **repeat-offender flag** (2+ warnings in 6 months, 3+ reporters in 90 days, or suspended before) and a workload table.
- **Hiding a message** keeps it for the record, tells the sender why, and can be undone.

- **Appeals** also cover warnings, hidden messages and dismissed reports. A **different moderator** must decide.
- Reporters follow each report and its outcome in Settings → My reports.

13 Accounts, security and privacy

Area	How it works
Sign-in	Email + password. The session token carries a password version, so a password change signs out other devices.
Password reset	By emailed code.
Social sign-in	Apple / Google tokens verified on the server (signature, issuer, audience, expiry). App buttons still need developer accounts.
Staff 2-step	Authenticator code (TOTP) plus 8 one-time recovery codes, required for staff routes.
Your data	Export everything; change email with a code; delete the account immediately.
Phone numbers	Stored encrypted (AES-256-GCM); a keyed hash answers "is this number already used?" without decrypting.
One person, many accounts	Max 3 accounts per device per 30 days; an indefinite suspension becomes a ban on the identity.
Identity check	Document-only method supported; provider still being decided.
Retention	A daily job deletes data past its retention period; an in-app screen explains how long things are kept.

The legal drafts in `docs/compliance/` carry [CONFIRM] marks and need a lawyer before launch.

14 The admin console

A separate web app for staff, with 2-step sign-in. Pages include reports, timed suspension, audit log, verifications, no-shows (moderator decisions), users (reliability panel, reset a person's 2-step) and Security.

15 Testing and demo data

Build, start and test from `services/api` with `npm run build`, `npm start` and `npm test`. **Never run two test runs at once**, they share one database. After editing route or error-code docs, run `npm run docs`.

Demo command	Creates
`npm run demo:feed`	Five demo people and 12 Milan activities
`npm run demo:circle`	Buddy, followed, follower and stranger activities and trips
`npm run demo:paging`	Lots of favorites, plans and alerts
`npm run demo:reliability`	Hosts and members with different histories
`npm run demo:waitlist`	A full activity with a waiting line

16 Where to find things in the code

I want to change…	Go to
Feed weights	`services/api/src/activities/feed-ranking.ts` (WEIGHTS)
What counts as a buddy	`services/api/src/common/social.ts`
Trip bonus for buddies	`services/api/src/trips/trips.service.ts` (search)
Waiting list	`activities.service.ts` → fillFromWaitlist
Cursor format and limits	`services/api/src/common/paging.ts`
304 / ETag behaviour	`services/api/src/main.ts` , `apps/mobile/src/lib/api.ts`
Reliability rules	`services/api/src/reliability/reliability.ts`
Database tables	`services/api/prisma/schema.prisma`
API list and error codes	`services/api/src/docs/`
The green "buddy going" line	`apps/mobile/src/components/Cards.tsx`

17 Known open items

- Map pins may flicker on iPhone (Expo Go) when closing the activity sheet; not confirmed fixed.
- Chat and host-request screens still poll quickly.
- The relevance feed applies filters after the 500-item pool; trip search re-scores on every page.
- No appeal route for a reliability level itself; no check yet for hosts who over-mark no-shows.
- Privacy notice, terms and consent screens are not in the app yet (need a lawyer first).
- Apple / Google buttons are not in the app; the identity provider is undecided.